

Calculus and Linear Algebra

Parametric Linear Systems

Lecture 9

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Lecture outline

- ▶ What changes when a coefficient or a constant depends on a parameter k .
- ▶ Generic values and exceptional values of k .
- ▶ Classification by determinant, rank and Gaussian elimination.
- ▶ Parametric 2×2 , 2×3 , 3×2 and 3×3 systems.
- ▶ Unique solutions, no solutions and infinitely many solutions in the same family.
- ▶ Homogeneous parametric systems and independent practice.

Aim

Given a family of systems depending on k , determine the complete solution set for **every** $k \in \mathbb{R}$.

What is a parametric linear system?

A parametric system is a family of linear systems whose coefficients or constants depend on a real number k .

$$A(k)x = b(k), \quad k \in \mathbb{R}.$$

For each fixed value of k , we obtain one ordinary linear system.

The question is not “find one solution”

We must classify the family: for which values of k is the system compatible, incompatible, uniquely solvable or infinitely solvable?

Do not confuse two different parameters

System parameter

$$\begin{cases} x + y = 2, \\ x + ky = 3. \end{cases}$$

The value of k changes the system itself.

Free solution variable

$$(x, y, z) = (2 - t, 1, t), \quad t \in \mathbb{R}.$$

The value of t selects one solution of a fixed system.

Throughout this lecture, $\boxed{k}$ is reserved for the **system parameter**. Free variables will be called t or s .

The classification principle

For every fixed k , the Rouché–Capelli theorem still applies:

$$\text{rank}(A(k)) < \text{rank}([A(k) \mid b(k)]) \implies \text{no solution.}$$

If the ranks are equal to r :

$$r = n \implies \text{one solution,}$$

$$r < n \implies \text{infinitely many solutions with } n - r \text{ free variables.}$$

The only new difficulty

The ranks may change when k takes special values.

A reliable workflow

For a system depending on k :

1. Write the coefficient and augmented matrices.
2. Perform elimination without dividing by expressions in k unless their non-zero condition is recorded.
3. Identify the values of k for which a pivot or determinant vanishes.
4. Solve the **generic case**.
5. Substitute every exceptional value into the reduced system or the original equations.
6. Compare the two ranks and solve each exceptional case separately.

Golden rule

Never cancel or divide by $k - a$ before separating the case $k = a$.

Square systems: determinant first

If $A(k)$ is square, the determinant immediately identifies the generic case.

$$\det A(k) \neq 0 \implies \text{rank}(A(k)) = n,$$

so the system has exactly one solution. If

$$\det A(k) = 0,$$

we only know that the coefficient matrix loses full rank.

Important

A zero determinant does **not** distinguish between no solution and infinitely many solutions. The augmented matrix must be checked.

A first 2×2 family

Exercise Classify for every $k \in \mathbb{R}$:

$$\begin{cases} x + y = 2, \\ x + ky = 3. \end{cases}$$

The coefficient matrix is

$$A(k) = \begin{pmatrix} 1 & 1 \\ 1 & k \end{pmatrix}, \quad \det A(k) = k - 1.$$

Thus $k = 1$ is the only exceptional value.

First family: the generic case $k \neq 1$

Subtract the first equation from the second:

$$\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 1 & k & 3 \end{array} \right) \sim \left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & k-1 & 1 \end{array} \right).$$

If $k \neq 1$,

$$y = \frac{1}{k-1}, \quad x = 2 - \frac{1}{k-1} = \frac{2k-3}{k-1}.$$

$$S_k = \left\{ \left(\frac{2k-3}{k-1}, \frac{1}{k-1} \right) \right\}, \quad k \neq 1.$$

First family: the exceptional case $k = 1$

For $k = 1$ the system becomes

$$\begin{cases} x + y = 2, \\ x + y = 3. \end{cases}$$

Elimination produces

$$\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 0 & 1 \end{array} \right).$$

Therefore

$$\text{rank}(A) = 1 < 2 = \text{rank}([A \mid b]).$$

$$\boxed{S_1 = \emptyset.}$$

The rational formulas from the generic case are not valid at $k = 1$.

Same coefficient matrix, different exceptional behaviour

Now consider

$$\begin{cases} x + y = 2, \\ x + ky = 2. \end{cases}$$

The determinant is again $k - 1$ and elimination gives

$$\left(\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & k-1 & 0 \end{array} \right).$$

If $k \neq 1$, then $y = 0$ and $x = 2$. The only question is again: what happens at $k = 1$?

Exceptional value with infinitely many solutions

If $k = 1$, both equations are identical:

$$x + y = 2.$$

Both ranks equal $1 < 2$, so there is one free variable. Set $y = t$. Then

$$x = 2 - t.$$

Hence

$$S_1 = \{(2 - t, t) : t \in \mathbb{R}\}.$$

Lesson

The same singular coefficient matrix can produce either no solution or infinitely many solutions, depending on the right-hand side.

Compare the two families

System	$k \neq 1$	$k = 1$
$\begin{cases} x + y = 2 \\ x + ky = 3 \end{cases}$	unique	no solution
$\begin{cases} x + y = 2 \\ x + ky = 2 \end{cases}$	unique	infinitely many

$$\det A(1) = 0$$

is only the beginning of the exceptional-case analysis.

One 2×2 family with all three outcomes

Exercise Classify

$$\begin{cases} kx + y = 2, \\ x + ky = 2. \end{cases}$$

The determinant is

$$D(k) = \det \begin{pmatrix} k & 1 \\ 1 & k \end{pmatrix} = k^2 - 1 = (k - 1)(k + 1).$$

Therefore the exceptional values are

$$k = 1 \quad \text{and} \quad k = -1.$$

All three outcomes: generic case

For $k \neq \pm 1$, the determinant is non-zero, so the solution is unique. Subtract the two equations:

$$(k - 1)(x - y) = 0.$$

Since $k \neq 1$, we get $x = y$. Then

$$(k + 1)x = 2.$$

Because $k \neq -1$,

$$x = y = \frac{2}{k + 1}.$$

Thus

$$S_k = \left\{ \left(\frac{2}{k + 1}, \frac{2}{k + 1} \right) \right\}, \quad k \neq \pm 1.$$

All three outcomes: $k = 1$

At $k = 1$ the system becomes

$$\begin{cases} x + y = 2, \\ x + y = 2. \end{cases}$$

The two equations coincide, so both ranks equal 1. Choose $y = t$:

$$S_1 = \{(2 - t, t) : t \in \mathbb{R}\}.$$

There are infinitely many solutions.

All three outcomes: $k = -1$

At $k = -1$:

$$\begin{cases} -x + y = 2, \\ x - y = 2. \end{cases}$$

Adding the equations gives

$$0 = 4,$$

a contradiction. Hence

$$\text{rank}(A) = 1 < 2 = \text{rank}([A \mid b]),$$

so

$$S_{-1} = \emptyset.$$

Complete classification of the 2×2 family

Value of k	Classification	Solution set
$k \neq \pm 1$	unique	$\left\{ \left(\frac{2}{k+1}, \frac{2}{k+1} \right) \right\}$
$k = 1$	infinitely many	$\{(2 - t, t) : t \in \mathbb{R}\}$
$k = -1$	no solution	$\emptyset$

This is the model answer

A parametric problem is complete only when every real value of k belongs to exactly one analysed case.

A 2×3 family: generic rank 2

Exercise Classify and solve

$$\begin{cases} x + y + z = 2, \\ 2x + ky + 2z = 4. \end{cases}$$

Eliminate x :

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & k & 2 & 4 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & k-2 & 0 & 0 \end{array} \right).$$

The exceptional value is $k = 2$.

2×3 family: $k \neq 2$

If $k \neq 2$, then

$$y = 0.$$

The first equation becomes

$$x + z = 2.$$

There are three unknowns and rank 2, hence one free variable. Set $z = t$:

$$S_k = \{(2 - t, 0, t) : t \in \mathbb{R}\}, \quad k \neq 2.$$

The solution set is a line in $\mathbb{R}^3$.

2×3 family: $k = 2$

If $k = 2$, the reduced matrix becomes

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right).$$

Both ranks equal 1. Since $n = 3$, there are two free variables. Set

$$y = s, \quad z = t.$$

Then

$$S_2 = \{(2 - s - t, s, t) : (s, t) \in \mathbb{R}^2\}.$$

The solution set has enlarged from a line to a plane.

Changing one constant changes the exceptional case

Now consider

$$\begin{cases} x + y + z = 2, \\ 2x + ky + 2z = 5. \end{cases}$$

Elimination gives

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & k-2 & 0 & 1 \end{array} \right).$$

If $k \neq 2$,

$$y = \frac{1}{k-2}.$$

Set $z = t$; then

$$x = 2 - \frac{1}{k-2} - t.$$

There is still one free variable.

The same 2×3 family at $k = 2$

When $k = 2$, the last row becomes

$$(0 \quad 0 \quad 0 \mid 1),$$

which means $0 = 1$. Therefore

$$\text{rank}(A) = 1 < 2 = \text{rank}([A \mid b]).$$

$$S_2 = \emptyset.$$

Rectangular systems

For 2×3 systems, the generic compatible case usually has one free variable. A rank drop may create two free variables or incompatibility.

A 3×2 family: compatibility is the issue

Exercise Classify

$$\begin{cases} x + y = 2, \\ 2x - y = 1, \\ 3x + ky = 4. \end{cases}$$

The first two equations already form a non-singular 2×2 system:

$$\det \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} = -3 \neq 0.$$

They force

$$x = 1, \quad y = 1.$$

The third equation must now be checked.

3×2 family: one compatible value

Substitute $(x, y) = (1, 1)$ into the third equation:

$$3 + k = 4.$$

Hence compatibility requires

$$k = 1.$$

Therefore

$$S_1 = \{(1, 1)\},$$

and for every $k \neq 1$,

$$S_k = \emptyset.$$

Why no infinite solutions?

The first two coefficient rows already give coefficient rank $2 = n$, so every compatible case is unique.

A 3×2 family where the rank drops

Consider

$$\begin{cases} x + y = 2, \\ 2x + 2y = 4, \\ x + ky = 1 + k. \end{cases}$$

Subtract the first equation from the third:

$$(k - 1)y = k - 1.$$

Thus $k = 1$ is exceptional. The second equation is always twice the first and gives no additional information.

3×2 rank-drop family: $k \neq 1$

If $k \neq 1$, divide by $k - 1$:

$$y = 1.$$

Then the first equation gives

$$x = 1.$$

Hence

$$S_k = \{(1, 1)\}, \quad k \neq 1.$$

Both ranks equal $2 = n$, so the solution is unique.

3×2 rank-drop family: $k = 1$

At $k = 1$, the third equation becomes

$$x + y = 2,$$

so all three equations are multiples or copies of the same equation. Both ranks equal $1 < 2$. Set $y = t$:

$$S_1 = \{(2 - t, t) : t \in \mathbb{R}\}.$$

The exceptional value reduces the rank and creates one free variable.

A 3×3 family with all three outcomes

Exercise Classify and solve

$$\begin{cases} x + y + z = 3, \\ x + (1 + k)y + z = 3 + k, \\ x + y + (k^2 - 1)z = 4. \end{cases}$$

Subtract the first equation from the second and third:

$$ky = k, \quad (k^2 - 2)z = 1.$$

The exceptional values are therefore

$$k = 0, \quad k = \sqrt{2}, \quad k = -\sqrt{2}.$$

The determinant confirms the exceptional values

The coefficient matrix is

$$A(k) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1+k & 1 \\ 1 & 1 & k^2-1 \end{pmatrix}.$$

Using the same row subtractions,

$$\det A(k) = \det \begin{pmatrix} 1 & 1 & 1 \\ 0 & k & 0 \\ 0 & 0 & k^2-2 \end{pmatrix} = k(k^2-2).$$

Thus the determinant vanishes exactly at

$$0, \pm\sqrt{2}.$$

3×3 family: generic case

Assume

$$k \notin \{0, \sqrt{2}, -\sqrt{2}\}.$$

Then

$$y = 1, \quad z = \frac{1}{k^2 - 2}.$$

From the first equation,

$$x = 3 - y - z = 2 - \frac{1}{k^2 - 2}.$$

Therefore

$$S_k = \left\{ \left(2 - \frac{1}{k^2 - 2}, 1, \frac{1}{k^2 - 2} \right) \right\}.$$

The solution is unique because the determinant is non-zero.

3×3 family: the exceptional value $k = 0$

For $k = 0$ the reduced equations are

$$0 = 0, \quad -2z = 1.$$

Thus

$$z = -\frac{1}{2}.$$

The first equation becomes

$$x + y = \frac{7}{2}.$$

Set $y = t$. Then

$$S_0 = \left\{ \left(\frac{7}{2} - t, t, -\frac{1}{2} \right) : t \in \mathbb{R} \right\}.$$

Both ranks equal $2 < 3$: infinitely many solutions with one free variable.

3×3 family: $k = \pm\sqrt{2}$

If $k = \pm\sqrt{2}$, then

$$k^2 - 2 = 0.$$

The third reduced equation becomes

$$0 \cdot z = 1,$$

that is,

$$0 = 1.$$

Hence the system is incompatible:

$$S_{\sqrt{2}} = S_{-\sqrt{2}} = \emptyset.$$

The first exceptional value $k = 0$ produced infinitely many solutions; the other two produce none.

Complete classification of the 3×3 family

Value of k	Classification	Solution set
$k \notin \{0, \pm\sqrt{2}\}$	unique	$\left\{ \left(2 - \frac{1}{k^2-2}, 1, \frac{1}{k^2-2} \right) \right\}$
$k = 0$	infinite	$\left\{ \left(\frac{7}{2} - t, t, -\frac{1}{2} \right) : t \in \mathbb{R} \right\}$
$k = \pm\sqrt{2}$	none	$\emptyset$

A singular 3×3 family for every k

Not every parametric problem has a generic non-zero determinant. Consider

$$\begin{cases} x + y + z = 3, \\ 2x + 2y + 2z = 6, \\ x + ky + z = 3. \end{cases}$$

The second equation is always twice the first, so

$$\det A(k) = 0 \quad \text{for every } k.$$

We must use rank and elimination instead of Cramer.

Singular family: $k \neq 1$

Subtract the first equation from the third:

$$(k - 1)y = 0.$$

If $k \neq 1$,

$$y = 0.$$

Then

$$x + z = 3.$$

Set $z = t$:

$$S_k = \{(3 - t, 0, t) : t \in \mathbb{R}\}, \quad k \neq 1.$$

The rank is $2 < 3$, so there is one free variable.

Singular family: $k = 1$

At $k = 1$, the third equation is also

$$x + y + z = 3.$$

The coefficient rank drops from 2 to 1. There are two free variables. Set

$$y = s, \quad z = t.$$

Then

$$S_1 = \{(3 - s - t, s, t) : (s, t) \in \mathbb{R}^2\}.$$

The exceptional value enlarges the solution set from a line to a plane.

Change the right-hand side: same rank drop, different outcome

Now change only the third constant:

$$\begin{cases} x + y + z = 3, \\ 2x + 2y + 2z = 6, \\ x + ky + z = 4. \end{cases}$$

Subtract the first equation from the third:

$$(k - 1)y = 1.$$

If $k \neq 1$,

$$y = \frac{1}{k - 1},$$

and one free variable remains. What happens at $k = 1$?

The exceptional value now gives no solution

At $k = 1$, the reduced equation is

$$0 = 1.$$

Therefore

$$S_1 = \emptyset.$$

For $k \neq 1$, choose $z = t$ and use the first equation:

$$x = 3 - \frac{1}{k-1} - t.$$

Hence

$$S_k = \left\{ \left(3 - \frac{1}{k-1} - t, \frac{1}{k-1}, t \right) : t \in \mathbb{R} \right\}.$$

Homogeneous parametric systems

Consider

$$\begin{cases} x + y = 0, \\ (k - 1)y + z = 0, \\ (k + 2)z = 0. \end{cases}$$

This is homogeneous, so it is compatible for every k because

$$(x, y, z) = (0, 0, 0)$$

is always a solution. The determinant is

$$(k - 1)(k + 2).$$

Thus the exceptional values are $k = 1$ and $k = -2$.

Homogeneous family: complete solution

If $k \neq 1, -2$, all pivots are non-zero, so

$$S_k = \{(0, 0, 0)\}.$$

If $k = 1$, then $z = 0$ and $x + y = 0$:

$$S_1 = \{(-t, t, 0) : t \in \mathbb{R}\}.$$

If $k = -2$, then $z = 3y$ and $x = -y$:

$$S_{-2} = \{(-t, t, 3t) : t \in \mathbb{R}\}.$$

A special feature of homogeneous families

For

$$A(k)x = 0,$$

we always have

$$\text{rank}(A(k)) = \text{rank}([A(k) \mid 0]).$$

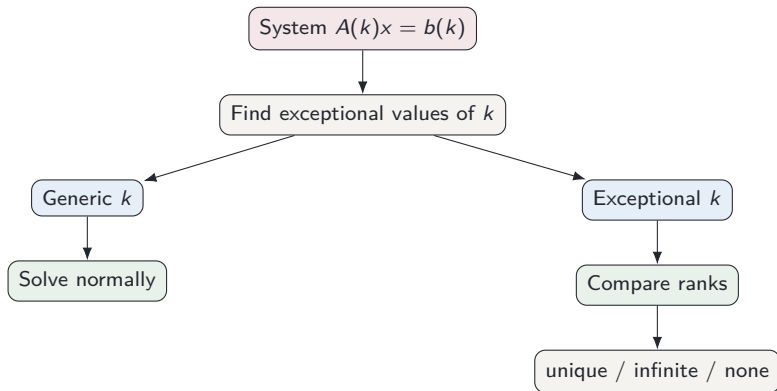
Therefore a homogeneous parametric system can never become incompatible.

- ▶ Full rank n : only the zero solution.
- ▶ Rank $r < n$: infinitely many solutions with $n - r$ free variables.

Exceptional values

In a homogeneous system, exceptional values can only increase the dimension of the solution space.

Decision tree for parametric systems



How to write the final answer clearly

A complete parametric answer should be organised by cases:

$$S_k = \begin{cases} \{\text{unique vector depending on } k\}, & k \in I_1, \\ \{\text{family depending on } t\}, & k = k_1, \\ \emptyset, & k = k_2. \end{cases}$$

Always state:

- ▶ the range of k for each formula;
- ▶ the number of solutions;
- ▶ the free-variable range $t \in \mathbb{R}$ or $(s, t) \in \mathbb{R}^2$ when needed.

Common mistakes

- ▶ Dividing by $k - a$ without first separating $k = a$.
- ▶ Using Cramer's rule at a value where $\det A(k) = 0$.
- ▶ Concluding “no solution” only because the determinant is zero.
- ▶ Confusing the system parameter k with a free solution variable.
- ▶ Solving only the generic case and forgetting exceptional values.
- ▶ Checking only $\text{rank}(A)$ and not the augmented rank.
- ▶ Forgetting that a rectangular system may be incompatible even when some square subsystem has a unique candidate.

Guided exercise A

Exercise Classify for every k :

$$\begin{cases} x + ky = 1, \\ kx + y = 1. \end{cases}$$

1. Compute the determinant.
2. Find the exceptional values.
3. Solve the generic case.
4. Analyse each exceptional value separately.

Do not use the generic formula at an exceptional value.

Guided exercise A: solution

Solution The determinant is

$$1 - k^2 = (1 - k)(1 + k),$$

so $k = \pm 1$ are exceptional. For $k \neq \pm 1$, symmetry gives $x = y$ and

$$(1 + k)x = 1,$$

so

$$x = y = \frac{1}{1 + k}.$$

At $k = 1$, both equations are $x + y = 1$:

$$S_1 = \{(1 - t, t) : t \in \mathbb{R}\}.$$

At $k = -1$, the equations are $x - y = 1$ and $-x + y = 1$, hence $0 = 2$:

$$S_{-1} = \emptyset.$$

Guided exercise B

Exercise Classify and solve

$$\begin{cases} x + y + z = 1, \\ 2x + 2y + 2z = 2, \\ x + ky + z = 1 + k. \end{cases}$$

Questions:

1. Which equation is always redundant?
2. What is the exceptional value of k ?
3. How many free variables occur in each case?
4. Give the complete solution sets.

Guided exercise B: solution

The second equation is twice the first. Subtract the first equation from the third:

$$(k - 1)y = k.$$

If $k \neq 1$,

$$y = \frac{k}{k - 1}.$$

Set $z = t$ and obtain

$$x = 1 - \frac{k}{k - 1} - t = -\frac{1}{k - 1} - t.$$

Thus

$$S_k = \left\{ \left(-\frac{1}{k - 1} - t, \frac{k}{k - 1}, t \right) : t \in \mathbb{R} \right\}, \quad k \neq 1.$$

At $k = 1$, the reduced equation is $0 = 1$, so

$$S_1 = \emptyset.$$

Independent practice

Classify each system for every $k \in \mathbb{R}$. **A**

$$\begin{cases} x + y = 3, \\ kx + y = k + 2. \end{cases}$$

B

$$\begin{cases} x + y + z = 2, \\ x + ky + z = 2. \end{cases}$$

C

$$\begin{cases} x + y = 2, \\ 2x - y = 1, \\ kx + 2y = 3. \end{cases}$$

For every case, state both ranks at the exceptional values and give the complete solution set.

Independent practice: answers

A: subtract the first equation from the second:

$$(k - 1)x = k - 1.$$

If $k \neq 1$, $x = 1$, $y = 2$. If $k = 1$, both equations are $x + y = 3$:

$$S_1 = \{(3 - t, t) : t \in \mathbb{R}\}.$$

B: subtract the first equation from the second:

$$(k - 1)y = 0.$$

If $k \neq 1$, $y = 0$ and

$$S_k = \{(2 - t, 0, t) : t \in \mathbb{R}\}.$$

If $k = 1$, one equation remains:

$$S_1 = \{(2 - s - t, s, t) : (s, t) \in \mathbb{R}^2\}.$$

Independent practice C: answer

The first two equations give

$$x = 1, \quad y = 1.$$

The third equation requires

$$k + 2 = 3,$$

so compatibility occurs only for

$$\boxed{k = 1.}$$

Therefore

$$\boxed{S_1 = \{(1, 1)\}}, \quad \boxed{S_k = \emptyset \text{ for } k \neq 1.}$$

For $k = 1$, both ranks equal 2. For $k \neq 1$, the coefficient rank is 2 and the augmented rank is 3.

Final checklist for an exam problem

Before finishing a parametric-system exercise, ask:

1. Did I find **all** exceptional values of k ?
2. Did I state the assumptions under which I divided by an expression involving k ?
3. Did I analyse each exceptional value separately?
4. Did I compare coefficient and augmented ranks when necessary?
5. If solutions are infinite, did I give **all** of them using t or (s, t) ?
6. Does every $k \in \mathbb{R}$ belong to exactly one final case?

Lecture 9 recap

- ▶ A parametric system is a family $A(k)x = b(k)$.
- ▶ Generic values are solved normally; exceptional values require separate analysis.
- ▶ For square systems, $\det A(k) \neq 0$ guarantees a unique solution.
- ▶ When the determinant or a pivot vanishes, compare the two ranks.
- ▶ Exceptional values may create infinitely many solutions or incompatibility.
- ▶ Rectangular systems require particular attention to compatibility.
- ▶ Homogeneous systems are always compatible.

End of the linear-systems unit

We now have the complete toolkit: elimination, rank, Rouché–Capelli, Cramer and parametric classification.